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Characterization-Guided Framework for Structure-Aware Graph Augmentation for GNNs

Research Master's Thesis Report - Data and Intelligence for Smart Systems (DISS)

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Characterization-Guided Two-Stage Framework for Structure Aware Graph Augmentation for GNNs

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Abstract

Structural-role graph augmentation adds virtual links between possibly far-away nodes that share similar node roles, enabling a Graph Neural Network to share information between role-similar nodes that do not have a direct connection in the original graph. This work introduces a pre-training decision framework to decide whether to apply the structural-role augmentation or not, and when the augmentation is suitable, then which structural signal to build the virtual graph. The first stage characterizes the graph to filter out the cases where the original topology should be preserved as compared to the cases where augmentation can help.

In the second stage, we introduce adjusted neighbour predictability as a pre-training criterion for deciding when centrality-based structural augmentation should be selected. This study measures this graph characteristic for centrality-based graph selection, providing an interpretable alternative to applying the same rewiring strategy across all datasets. The details can be accessed here: <https://github.com/SafiUllahAdam/Structure-Aware-Graph-Augmentation-for-GNNs>

1 Introduction

A Graph Neural Network (GNN) [1], is a type of machine learning model used for graph-structured data. A GNN model considers the information of the Neighbouring nodes and connected nodes around the node being viewed, instead of just the node alone. A simple way to imagine this is a person asking their neighbours for guidance or information before making an informed decision. By repeating this process, a node can learn from both its own information and its surrounding connections. We used GraphSAGE [2] in this research. GraphSAGE is a class of GNN where the node representation gets learned from its local neighbourhood. It is inductive in nature, which means it learns how to create node representations based on neighbourhood information and can therefore be applied to previously unseen nodes as well. It doesn't necessarily need to use all of the neighbours; it could sample a set of neighbours, and then add them up, then aggregate their information into one useful representation.

The connections in a graph can strongly affect what a learning model is able to understand. Although some nodes have similar structure in the graph, it does not mean that they are directly connected. This research focuses on graph augmentation through edge generation, i.e. addition of carefully selected virtual edges in order to produce additional connections. The main idea is not to add edges blindly, but to use properties of the graph to help decide which augmentation strategy is more suitable.

1.1 Problem Statement

In Graph Neural Networks (GNNs) the information shared between connected nodes during the learning process is used to build nodes representations. This is effective if information in the neighbourhoods given by the original graph is useful to the downstream task. A common limitation of message-passing GNNs is that their performance can depend strongly on whether the graph neighbourhoods contain useful information for the task, particularly when the graph is heterophilous or when repeated aggregation causes excessive mixing of node representations. Specific message passing GNNs have shown enhanced performance when the labels or features of connected nodes are similar, more commonly referred to as the homophily property. If there is a difference in nodes connected together, that is, heterophily, the original neighbourhood may not be a good source of information for learning [3]. Several GNN layers can be passed multiple times, causing the representations of nodes to become more similar, which is called over-smoothing.

A second limitation is that being in the same location does not necessarily mean that it is the same structure. In a network, similar structural functions can be assumed by different nodes, even if they are not directly connected and are situated far away from each other, for example hub, bridge, or peripheral node. To solve this issue, the concept of “structural similarity,” “guided random walk,” and “Skip-Gram representation learning” was used in the Identity2Vec [4]. For this research, we directly encode such structural similarities as virtual graph connections and have GraphSAGE run the message passing process across them.

Changing the original graph topology with structural-role connections poses the main challenge as it might benefit one graph but not the other; thus, it must be determined whether this augmentation enhances GNN learning or not. In this problem, we explore this challenge from two perspectives that are performed as downstream tasks: the **node classification** and the **link prediction** tasks. Most of the rules developed for the characterization step in this study are based on the link-prediction experiments.

1.2 Significance

The problem that arises here then is not only how to form the structural-role connections but whether the original graph should be modified or not. The virtual edge alters the Neighbourhoods in which a GNN spreads information, meaning that an augmentation that benefits one graph can be detrimental to another. This concern has been substantiated by research on characterizing graphs which revealed that the properties of graphs are not homogeneously interpretable across datasets. “Homophily” refers to the tendency of Neighbouring nodes in the network to belong from the same class; for instance, friends have similar interests and thus belongs to same friends circle. In this research, we have used refined version called adjusted homophily: by correcting for similarities that could happen simply because some classes are more common than others, and thus provides a fairer measure of the strength of the graph’s homophily. Specifically, the adjusted homophily was suggested to address the shortcomings of the existing homophily measures, and the information contained in labels of Neighbourhoods can be used in the graph behavior analysis [5].

In this research, characterization means measuring properties at the graph-level prior to taking the ultimate decision made by GNN and associating those properties with outcomes observed in the original-augmented graph gap. So it is a diagnostic stage, rather than an embedding technique. Adjusted homophily, connectivity statistics, Neighbour predictability, and the project-defined original-edge-retention statistic are analyzed to determine the circumstances under which one would want to retain the observed topology or want to think about structuring an augmentation. We then freeze the resulting decision thresholds and run them on held out/ validation graphs.

Graph → Characterize it → Compare graph properties with results
→ Discover a rule → Apply the rule to an unseen graph for validation

1.3 Proposed Solution

Therefore in this study, we propose a two-stage pre-training decision methodology. First, characteristics of the graph are analyzed for retention of the original topology or structurally augmenting the graph. An important signal is that high adjusted homophily is always beneficial for maintaining the original graph, whereas low homophily is not enough to ensure augmentation benefits. Second, if augmentation is warranted, the system determines which structural signal should construct the additional role edges. The introduced “Stage-2” rule employs an adjusted neighbour predictability: above this threshold (around 0.0092) eigenvector-centrality based augmentation is preferred, while below this value, centrality is not necessarily the best signal from the others, e.g., the degree or Ψ .

Many popular GNN models like GraphSAGE actually fails to distinguish certain simple graph structure, which means they cannot see the structural similarity between any 2 nodes in graphs if there are not any physical edges present between them, which brings us to this research. We propose a whole spectrum of different graph variants to look at the unchanged original graph or pure role-based virtual graphs and to look at the hybrid variants which includes both original edges and virtual edges. Following figure 1 shows the names and details of variants.

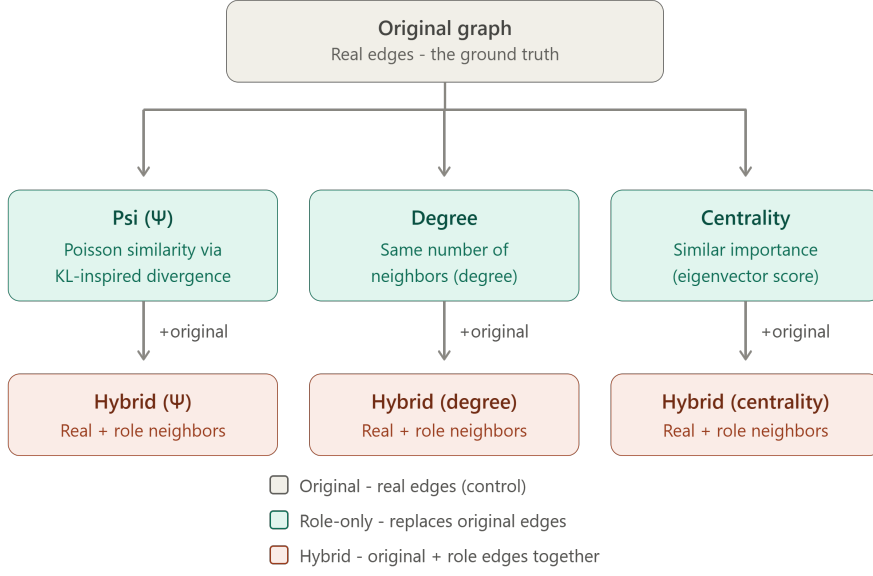


Figure 1: Graph Types

- **Ψ Structural Score Graph**

In the Ψ -based virtual graph, each node is linked to the top- K nodes that have the most similar structural pattern, based on their Ψ score, which has been explained in detail in Section 3.3.

- **Degree Graph**

In this graph, nodes get connected based upon similar number of neighbours. It demonstrates whether only degree is enough to identify the nodes with similar structural properties.

- **Centrality Graph**

In this graph, we calculated eigenvector centrality not simple centrality. Eigenvector centrality assigns to a node a score based on the number of nodes with high importance that are connected to it. The centrality graph is constructed by making each node linking to the top- K most similar eigenvector-centrality values. It is useful to test whether global importance of a node in graph is useful for constructing role-based connections.

- **Original Graph**

This virtual graph uses dataset's original edges without adding or replacing anything. It acts as a baseline control which is useful to compare whether the virtual graphs are helping in real.

- **Hybrid Graphs**

Combines original edges along with additional Ψ structural similarity based edges. In this manner, GNN can utilize both the original edges and additional structural-based connections being added.

- **Hybrid Graph Degree**

The Hybrid Degree graph keeps all original edges and adds new role-based edges between nodes with similar degree values. This allows nodes with a similar number of neighbours to exchange information while still preserving the original graph structure.

- **Hybrid Graph Centrality**

The Hybrid Centrality graph keeps all original edges and adds new role-based edges between nodes with similar eigenvector-centrality values. This allows nodes with similar structural importance in the graph to become connected without removing their original relationships.

A static top- K selection process is used for the virtual graph construction. First, all the values of the nodes, such as their degree, Ψ value or their eigenvector centrality are computed from the original graph and frozen at this step. Every node then chooses the $K = 10$ most similar nodes, using these frozen values, and all of the connections added to a new virtual graph. New edges are not going to either change the value of nodes or influence the subsequent selections. The process is hence not ‘iterative rewiring’. If the similarity values for multiple nodes are equal, then the tied nodes are selected uniformly at random from a random seed, which allows us to construct graphs in a reproducible way. The final degree of a node can be larger than K in some cases because other nodes can make their own selections of the node that make it K degrees.

2 Literature Review

Graph neural networks (GNNs) learn by exchanging information through graph edges, meaning that the topology supplied to a GNN directly controls which nodes can influence one another. Consequently, changing the graph is not merely a preprocessing choice; it changes the information available to message passing. Prior work shows that observed graphs may be noisy, incomplete, or poorly aligned with a downstream task, motivating graph structure learning and augmentation [6]. At the same time, structural-role research shows that nodes can play similar roles even when they are far apart and not directly connected [7] [8].

2.1 Graph Characterization and Heterophily

The strongest foundation for Stage 1 comes from heterophily research, although this literature also shows why homophily should not be treated as a complete decision variable, because even graphs with low homophily can perform well when the model captures useful higher-order neighbourhood information, [3] demonstrated that widely used message-passing GNNs can perform poorly when connected nodes have different labels. Their H2GCN model addressed this through higher-order Neighbourhoods and combinations of intermediate representations. The study established that the relationship between labels and topology matters strongly to GNN behavior, but its solution changes the GNN architecture rather than deciding whether the input topology should be preserved or structurally augmented [3].

Another study [5] moved considerably closer to the objective of the present study by asking how graphs themselves should be characterized. They showed that several common homophily measures have weaknesses that make cross-dataset comparisons difficult and found that adjusted homophily satisfies more desirable measurement properties. More importantly, they proposed label informativeness which captures how much information about a node’s label being provided by a Neighbour’s label, and found that it agrees with GNN performance better than homophily measures in their experiments [5]. This literature provides a direct justification for using adjusted homophily as a Stage-1 characterization measure, while simultaneously warning that low homophily alone cannot explain every graph as we have seen in results of our experiments which has been discussed in section 4.2.

Later evidence reinforces this point [9]. It Separated graph homophily into label, structural, and feature aspects and reported that their combined Tri-Hom measure correlated with model performance better than 17 single-aspect measures over 31 real-world datasets [9]. Similarly, they found that the relationship between heterophily level and the most suitable graph filtering behavior is not strictly monotonic. Heterophily means the inclination of nodes which are connected, to posses different labels means a node’s immediate connected node may not necessarily belongs to same class and thus provides similar information. These studies do not weaken the case for adjusted homophily; rather, they suggest that it is better viewed as a screening variable within a larger decision process [10], which is consistent with the current project’s additional retention gate for low-homophily graphs.

SHGNN study [11] introduced a Spatial Diversity Score to measure how heterophily varies across spatial-distance groups in urban graphs and designed a specialized architecture to process these spatial patterns. Experiments were conducted on three urban datasets. The study confirms that characterizing heterophily can guide GNN design [11], but it does not duplicate the present research question: SHGNN studies spatial heterophily, uses a spatial metric, and proposes a new architecture, whereas the present study asks different questions.

2.2 Structural-role Representations and Virtual Graphs

A second research stream establishes why structural-role augmentation is meaningful. Struc2vec has been introduced to represent nodes according to structural identity rather than simple network proximity. It measures structural similarity between nodes at several scales, constructs a multilayer similarity graph, and generates contexts in which structurally equivalent nodes can become close in the learned representation. GraphWave [8] similarly represents structural roles through heat-wavelet diffusion patterns and shows that nodes in distant parts of a network can have similar structural representations when their local topologies are alike.

Identity2Vec (I2V) is particularly important to the current project. It targets mesoscopic structural identity, using a Poisson similarity signature value, divergence estimation, guided walks, and Skip-gram [12] to learn embeddings for structurally similar nodes. The collective contribution of this literature is clear: adjacency and structural equivalence encode different information.

However, these methods answer a different question. They primarily ask how structural identity should be represented after structural similarity has already been chosen as the objective. They do not give a graph-level rule for deciding whether the original edges should remain dominant, whether role-based edges should be added or substituted, or whether Ψ , degree, or centrality is the appropriate signal for an unseen dataset. Therefore, neither structural identity nor Centrality itself constitutes the novel part of the present work. The research opportunity lies in deciding when structural-role information should alter GNN connectivity and which form of structural identity should be used.

2.3 From Graph Augmentation to Adaptive Topology Selection

Graph augmentation and structure-learning research is the closest competing literature. For instance, [6] jointly learned a distribution of discrete probability over edges and GCN parameters through bilevel optimization, allowing graph structure to be learned when the observed topology is incomplete, corrupted, or absent. And [13] proposed GAUG, which uses neural edge prediction to promote class-homophilic structure for semi-supervised node classification. Their experiments showed improvements across several architectures and datasets. Another study approached heterophily through graph rewiring [14], adding edges similar in homophilic nature and pruning likely heterophilic ones based on Neighbourhood feature distributions or label, evaluated on 11 datasets.

These studies establish that changing graph connectivity can improve GNN learning, but they normally proceed directly toward a learned or heterophily-reduced topology. Their principal question is how to improve the graph, rather than first testing whether changing it is preferable to keeping the observed topology. They are also mainly developed around node classification, whereas the retained characterization rules in the present project arose from link-prediction analysis.

The closest study for the strategy-selection part is AutoGDA [15]. It explicitly argued that applying one augmentation strategy uniformly can be suboptimal because communities can differ in properties such as degree distribution and homophily. AutoGDA formulates augmentation selection as bilevel optimization and uses reinforcement learning to learn community-specific augmentation strategies for node classification, reporting gains of up to 12.5 percentage points over its comparison methods. Similarly [15] co-optimizes input graph topology and GNN architecture through differentiable bilevel search, particularly targeting heterophilic settings.

Rather than searching augmentation policies through repeated model optimization, this study aims to derive an interpretable graph-level decision from measurable graph characteristics "before" GNN training.

2.4 Research Gap

Graph-characterization research further supports this direction, [16] analyzed GNN benchmark meta-data using multivariate sparse regression to identify graph properties associated with performance and then investigated degree distribution theoretically and experimentally. Their results demonstrate that dataset properties can reveal why GNNs behave differently across graphs. However, their framework [16] remains primarily explanatory: it does not transform graph characterization into an executable keep-original versus augment and then to choose-augmentation-signal pipeline which we are doing in this study.

The literature therefore leaves a clear intersection insufficiently addressed. Existing studies generally characterize graph conditions without prescribing structural-role augmentation, construct structural-role representations without deciding when they are needed, or optimize topology during training rather than deriving a simple pre-training decision rule. The present study connects these areas. Accordingly, the strongest thesis claim is not that homophily matters, that structural roles are useful, or that augmentation strategies can be selected automatically; all three ideas have prior literature. Therefor, the research questions we are trying to find here are:

- RQ1: The first research question is when is a graph’s topology sufficient for a GNN and when does introducing structural role-based connections boost the performance?
- RQ2: What structural signal of graph Ψ , degree or eigenvector centrality should be employed to establish the connecting structures, in case of an augmentation?

The more specific contribution is the formulation and empirical testing of a Characterization-Guided Two-Stage Decision Framework for Structural Role-Graph Augmentation: first determine whether structural augmentation is warranted, then determine which role signal should construct it. This framing is very similar to the study goals while still leaving a meaningful extension to the research paper as the study could go forward to develop and evaluate additional Stage-2 rules to separate out Ψ and Degree, and determine if these decisions carry over to other datasets, tasks, encoders, and augmentation densities.

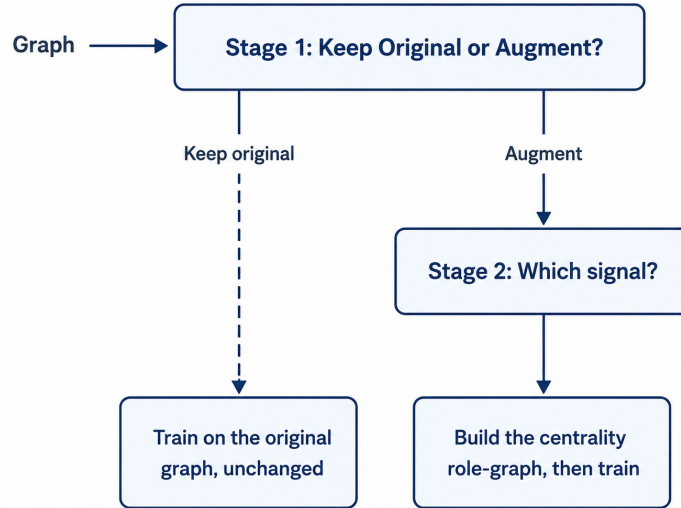


Figure 2: Architecture Diagram

3 Methodology

3.1 Research Design and Scope

In this study, we followed the design which is quantitative, controlled experimental and comparative. The graph construction and structural signals were systematically changed in the evaluation of multiple graph datasets while keeping the same GraphSAGE parameters. There was a comparison of performance for downstream tasks such as node classification and link-prediction, followed by graph-property characterization, rule fitting, ablation analysis and held-out validation for a two-stage augmentation decision framework. This design was chosen based upon the primary goals of study. Holding the encoder and evaluation procedure fixed while changing the graph supplied a controlled way to attribute performance differences to graph structure rather than to changes in model architecture or optimization.

The research was conducted entirely as a computational benchmarking study using publicly available graph datasets. All experiments were organized in a Python research repository containing fixed dataset registries, graph-building procedures, evaluation scripts, saved splits, and result tables. The analysis deliberately treated the graphs as structural-only inputs for the main framework. For datasets that originally contained attributes, including ogbn-arxiv, Roman-empire, PubMed, Actor, Amazon Photo, and others, the original node features were excluded so that the effect of structural augmentation could be studied without being mixed with dataset-specific feature information. The dataset definitions, sources and this structural-only preprocessing policy are documented in the project repository.¹

3.2 Our Pipeline

The key goal of this project is to learn the node representation of a graph which is based on its structural identity, not node attributes or labels. The main difference between the two methods is how the structural information is converted into embeddings.

Virtual Role Graph for Structural Role Embeddings: While keeping the principle of preserving the structural-identity in I2V [4], we altered the way structural information is transformed into node embeddings. The degree-based information and eigenvector centrality are computed and stored once for the input graph, which means properties that are not changing for a fixed graph are not recalculated. Next, the structural signal [4] is transformed into a fixed structural signature. We do not rely on this signal to provide a hint for node walks but to build a virtual graph where nodes are linked based on structural similarity. The graph’s edges are not only the connections nodes record as they were made from the beginning; they are also the edges between nodes that are structurally similar, with the top- K most similar nodes connected to each node.

The virtual graph is then provided to GraphSAGE, which learns representations of the nodes by taking information from the neighbouring nodes in the virtual graph. So the essential features of I2V [4] that have been adopted are: the structural properties and the structural-identity principle that relies on the KL/Poisson properties. The main components replaced are the probability-guided walks and Skip-gram embedding model, which are replaced by a virtual graph and GraphSAGE, respectively. So the pipeline is

$$\begin{aligned} G &\rightarrow \text{Cached structural properties} \rightarrow \text{Structural similarity} \\ &\rightarrow \text{Virtual graph} \rightarrow \text{GNN} - \text{GraphSAGE} \rightarrow \text{Node embeddings} \end{aligned}$$

3.3 Structural Signal and Similarity

The virtual graphs built are based on the input graph’s structure information. Its goal is to explain the structural function of every node not using node attributes or class labels. This research maintains the key structural quantities defined in the Identity2Vec, such as degree distribution Δ , eigenvector centrality Ω , the KL-style divergence rate λ , and the Poisson identity score Ψ , but transforms the initial walk dependent score to a fixed point level structural signature. Write the input graph as

¹<https://github.com/SafiUllahAdam/Structure-Aware-Graph-Augmentation-for-GNNs>

$$G = (V, E), \quad (1)$$

Let V be the set of nodes, E the set of edges and let $\mathcal{N}(u)$ be the set of neighbours of the node u .

$$\deg(u) = |\mathcal{N}(u)|. \quad (2)$$

- **Degree Distribution Δ :** The degree-distribution value Δ_u shows how common a node's degree is in the graph by measuring the proportion of nodes that have the same number of neighbours as node u . We took the value of the degree distribution corresponding to the degree of each node. Suppose that there are $n = |V|$ nodes in total and n_d nodes with degree d . The degree-distribution value of a node u of degree d is

$$\Delta_u = \frac{n_{\deg(u)}}{n}, \quad (3)$$

Nodes with the same degree have the same value of Δ . Therefore, Δ_u shows how many nodes in the graph have the same degree as node u , relative to the total number of nodes.

- **Eigenvector centrality Ω :**

In a graph, the importance or influence of a node can be captured by centrality measures. Centrality measures can be different and they have varying definitions of importance, whether it be in relation to many connections, or to being connected to those who have most important connections. In this work we adopted eigenvector centrality that attributes greater importance to nodes that are connected to other highly important nodes. It thus reflects the number and quality of the connections that a node has.

Eigenvector centrality is based on the importance of neighbouring nodes.

$$\Omega_u = \frac{1}{c} \sum_{v \in \mathcal{N}(u)} A_{u,v} \Omega_v, \quad (4)$$

c is a scalar coefficient and $A_{u,v}$, the element in row u and column v of the adjacency matrix.

We performed eigenvector centrality analyses on each connected graph component in disconnected graphs. Let $C(u)$ be the connected component of the graph containing node u and let $\tilde{\Omega}_u$ be the eigenvector-centrality of node u within that component. For each non-trivial component, ViRGo normalizes the values to be

$$\Omega_u = \frac{\tilde{\Omega}_u}{\max_{z \in C(u)} \tilde{\Omega}_z}. \quad (5)$$

The maximum eigenvector-centrality value within each connected component is therefore equal to 1. If a node is isolated, then Ω_u is given the value of zero. This component-wise strategy prevents the measurement of centrality being influenced by other components that are not connected.

- **KL-style structural divergence λ :**

Each node in the KL-style structural divergence has one summary value λ_v which summarizes the structural pattern of its neighbourhood. In brief, it investigates the properties of the neighbourhood $N(x)$ of a node x and then compares two structural properties of each of its neighbours, their degree-distribution value Δ_x and their eigenvector centrality Ω_x . Such differences are summed up to one value, λ_v . This resulted in a structural quantity that was fixed, i.e. it was calculated once for each node and subsequently used to create the virtual graph. Given a node v the raw structural difference is computed as

$$R_v = \sum_{x \in \mathcal{N}(v)} \Delta_x \log \left(\frac{\Delta_x}{\Omega_x} \right), \quad (6)$$

where $\mathcal{N}(v)$ denotes the set of neighbours of node v . For each neighbour x , the degree-distribution value Δ_x is compared with its eigenvector-centrality value Ω_x . Only terms for which $\Delta_x > 0$ and $\Omega_x > 0$ are included in the calculation. To normalize the raw structural divergence R_v , we define a node-specific normalization term using the degree and eigenvector centrality of node v :

$$\omega_v = \deg(v) + \Omega_v. \quad (7)$$

Here, ω_v is an implementation-defined normalization term that combines the degree and eigenvector-centrality value of node v not a standard normalization. It is used to scale the raw divergence R_v before obtaining the final structural divergence rate λ_v . It is important to note that the combination of degree distribution and eigenvector centrality is based on the original Identity2Vec definition, and not as a novel normalization design in our project: these two quantities describe different aspects of the graph structure, and their sum has no theoretical normalization justification.

$$\lambda_v = \max \left(\frac{R_v}{\omega_v}, 10^{-12} \right). \quad (8)$$

The small lower bound of 10^{-12} is used only to avoid a zero or invalid value in the later Poisson calculation. Therefore, λ_v becomes one fixed structural value for each node, summarizing the structural pattern of its neighbourhood.

- **Poisson structural signature Ψ :**

The Poisson step converts the divergence rate λ_v into one structural signature for each node. In simple terms, λ_v tells us about the structural behaviour of the node's neighbourhood, while k_v tells us how many neighbours the node has. The Poisson formula combines these two values into a single score, making it easier to compare nodes with each other.

For node v ,

$$k_v = |\mathcal{N}(v)|$$

is the number of its neighbours. Using k_v and the previously calculated divergence rate λ_v , the Poisson score is

$$\Psi_v = \frac{\lambda_v^{k_v} e^{-\lambda_v}}{k_v!}. \quad (9)$$

In the implementation, this score is calculated in logarithmic form for numerical stability. The resulting value becomes the fixed structural signature of the node. Nodes with similar signature values are considered structurally similar, and these values are later used to select the top- K structural neighbours when building the virtual graph.

$$s_v = k_v \log(\lambda_v) - \lambda_v - \log(k_v + 1), \quad (10)$$

Log being used here to stabilize very small values. This value, s_v , was considered to be the fixed signature for a node to be compared against. New signature values are set to zero for isolated nodes (those with no neighbours).

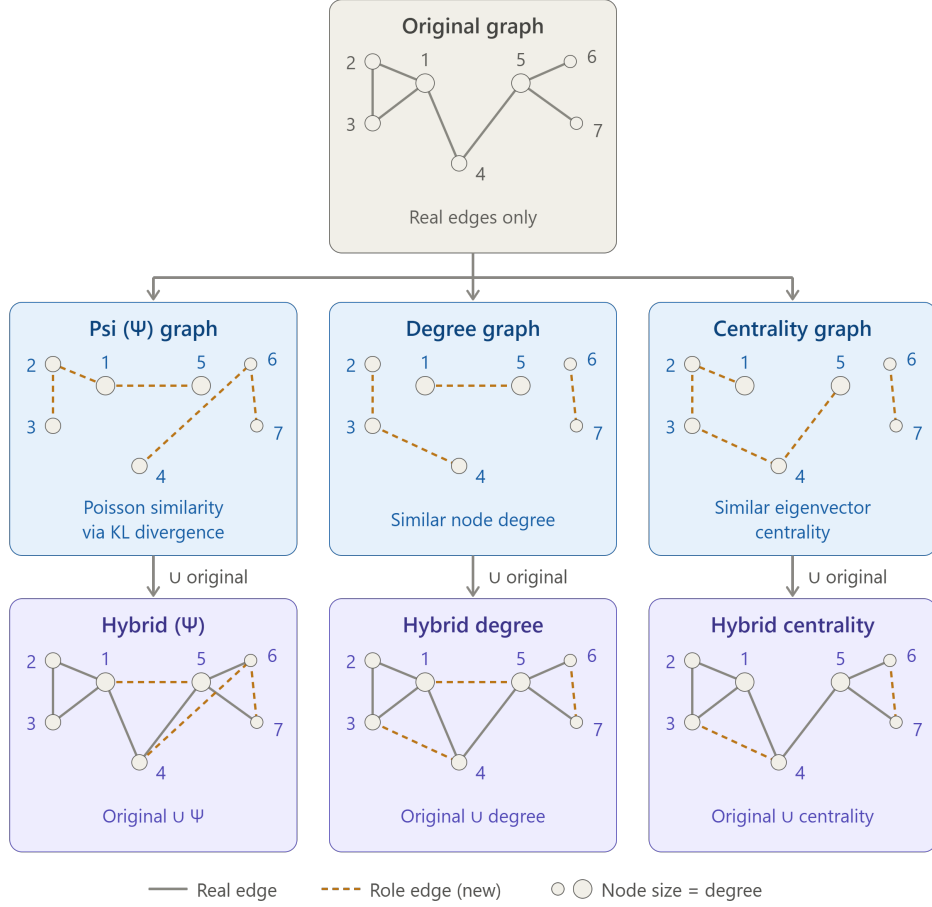


Figure 3: Graph Variants with Augmentation

- **Similarity of structure of nodes:**

Following the Poisson calculation, there is one fixed structural signature s_v at each node. As we are interested in seeing whether two nodes have a similar structural role between them, we compare their signatures using the absolute difference between them.

$$D_{\Psi}(u, v) = |s_u - s_v|. \quad (11)$$

A smaller value of $D_{\Psi}(u, v)$ indicates that the structural profile of nodes u and v were more similar and vice versa. For each node, we therefore select the $K = 10$ nodes with the smallest signature differences and connect them. These selected connections form the Ψ structural-role graph. A similarity weight w_{uv} is also assigned to each edge that is selected.

$$w_{uv} = \frac{1}{1 + D_{\Psi}(u, v)}. \quad (12)$$

Here, w_{uv} is a measure of the structural similarity of nodes u and v . The smaller the distance is, the closer the weight will be to 1, whereas a larger distance produces a smaller weight value w_{uv} .

3.4 Datasets, Structural Signals, and Model Configuration

The experimental pool covered citation, biological, social, interaction, co-purchase, heterophilous, and Facebook100 networks. It includes datasets in figure 4, not every dataset was used for both

tasks because some graphs did not contain aligned node labels. The two OGB datasets retained their official evaluation protocols, whereas the remaining applicable datasets followed the project’s common evaluation scheme. The use of graphs from different domains and with different homophily and connectivity patterns was appropriate because the proposed decision rules were intended to depend on graph properties rather than on a single application domain.

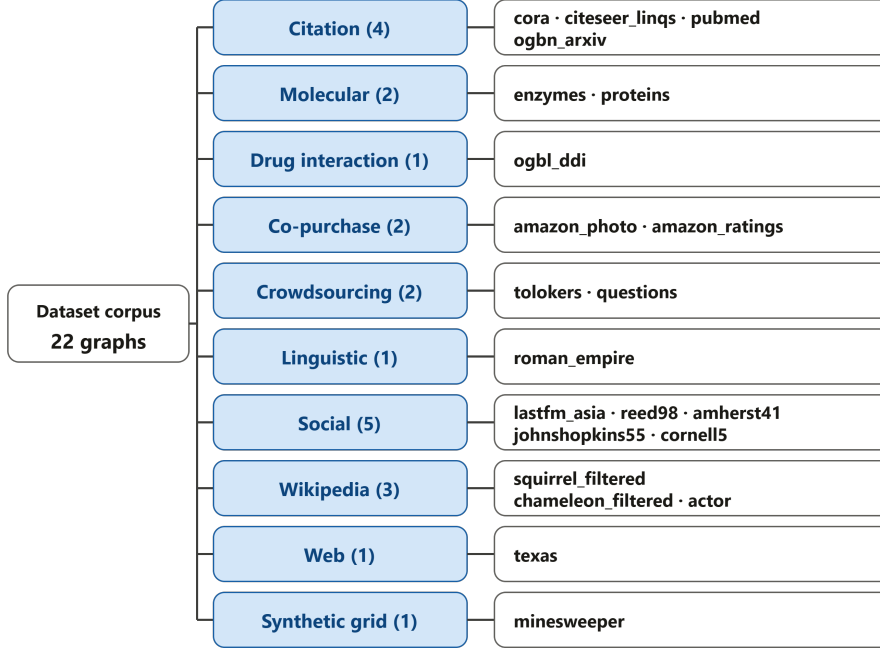


Figure 4: Dataset Corpus

Five principal graph conditions were initially treated as the locked comparison set: the unchanged original graph, an Identity2Vec-derived structural-role graph denoted Ψ , a degree-similarity graph, an eigenvector-centrality graph, and a hybrid graph combining original and role-based connectivity. Subsequent experiments also considered hybrid degree and hybrid centrality variants, while density-matched original-K and random-K graphs served as controls against the possibility that improvements arose merely from changing edge counts. Virtual Neighbourhoods were examined at K=5,10, and 20 out of which K=10 was finalized for subsequent experiments. Identity2Vec embeddings used 64 dimensions, walk length 40, 10 walks per node, context window 10, and one training epoch.

GraphSAGE was selected as the common message-passing encoder because it provided a standard Neighbourhood-aggregation architecture in which changing graph connectivity directly changed the Neighbourhoods from which representations were learned [2]. The fixed encoder contained two layers with mean aggregation, a 64-dimensional hidden representation and 64-dimensional output, a learning rate of 0.01, 50 training epochs, and five negative samples. It used a structural input representation that included eigenvector centrality, Identity2Vec’s Ψ and local clustering information, with the addition of degree. To maintain a constant graph topology as the experimental variable, these settings were left unchanged for all the different graph constructions used in the experiments.

3.5 Experimental Procedure and Decision Framework

The process was carried out in the order of representation construction, controlled downstream evaluation. All datasets were originally represented in the common representation which the repository has defined for graphs. Directed source networks were documented as such, but processed as per the undirected graph policy used in the project (where applicable). Then, using those nodes represented as vectors in the weights space, Identity2Vec representations of those nodes were computed, and then

alternatively, a different virtual graph with the predefined number $K=10$ of the most similar adjacent nodes was built. After experimenting with a few different values, the value $K=10$ was chosen as the size of the fixed neighbourhood for each node, as it provides sufficient structural connections to pass the messages while not creating too much of "noise" in the overall graph. Next, the fixed GraphSAGE encoder was then called with the same dataset, graph condition and experimental seed. This combination allowed for comparisons between the original and the augmented graph to be made under the same learning conditions. In the final experimental protocol, repeated measurements were combined for seeds 42, 43 and 44 as the default setting.

Stratification was employed to select 70% of nodes for the core node-classification protocol. Link prediction was run with a 70/30 edge split and the main unsupervised link score used was cosine similarity. There was no validation set since the study does not do hyperparameter tuning, all configurations were fixed prior to using a held-out dataset evaluation. The OGB treatment of the cases [17] is kept distinct to maintain the given benchmark definitions: In OGB-ARN, all the time series were kept in their original temporal split, with the same Accuracy measure was used; And in OGB-DD-I, only the training edges were used to build the graph while the following benchmark definitions were kept: in the evaluation set, only the positive and negative pairs were used from the official definition; and the same Hits@20 measure was employed. Hits@20 is an evaluation metric used to evaluate link prediction performance. It checks how often the missing link appears correctly among the top 20 predicted link by the model. A higher Hits@20 means the model correctly predicts the true connections among nodes.

After the graph-condition sweep, properties of the original graphs were characterized and related to the observed augmentation gaps. Candidate rules were fitted only on designated development datasets and were frozen before their corresponding held-out evaluations. The first-stage augmentation gate used adjusted homophily, with the locked threshold set at 0.227, followed in the refined gate by the repository-defined original-retention statistic with a threshold of 0.0119. The second stage addressed which structural augmentation should be chosen once augmentation had been indicated. It used adjusted Neighbour predictability, with a locked threshold of 0.0092 for the centrality-selection rule. Separate held-out graphs were subsequently passed through the frozen rules without refitting their thresholds. This separation between rule discovery and held-out application was important because it reduced direct adaptation of the final decision framework to the datasets on which its validity was assessed.

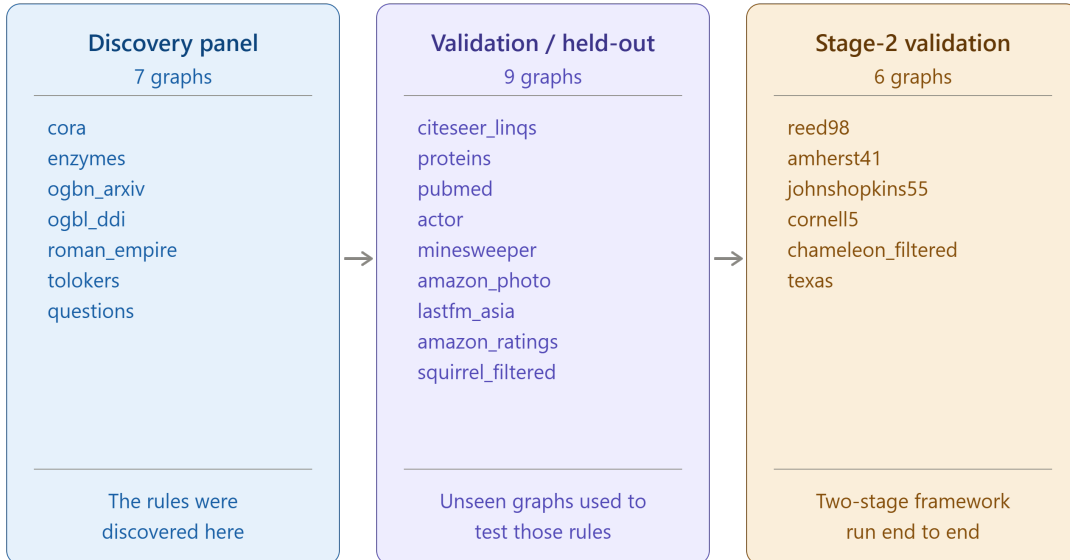


Figure 5: Dataset Groups in Stages

3.6 Outcome Measures

The primary outcomes were downstream predictive performance relative to the unchanged original-graph condition. Core node classification was evaluated using weighted F1, while core link prediction was evaluated using ROC-AUC; the OGB datasets retained their benchmark-specific Accuracy and Hits@20 outcomes. These measures were appropriate because the study examined both node-level discrimination and the recovery of graph relationships rather than relying on a single task.

Secondary outcomes included the performance difference between each augmented condition and the original graph, variability across random seeds, structural graph diagnostics, original-edge retention, and the agreement between frozen rule recommendations and the preferred graph condition. Graph-level characteristics were also used as explanatory variables to determine whether inexpensive properties of an unseen graph could predict augmentation behavior without requiring a full augmented-model sweep.

3.7 Statistical Analysis

Repeated-run performance was summarized using the mean, standard deviation, and standard error of the mean across experimental seeds. Graph-level associations between candidate diagnostic variables and augmentation behavior were examined using Spearman rank correlation, which was suitable for the relatively small heterogeneous graph panel and did not require a linear relationship or normally distributed graph statistics. The characterization workflow incorporated Bonferroni adjustment when multiple prespecified predictors were screened, using a familywise significance level of $\alpha=.05$. Threshold-based decision rules were then treated as predictive rule-fitting procedures rather than as conventional null-hypothesis significance tests. Candidate cut points were derived from observed graph-level predictor separations, locked on the development panel, and subsequently evaluated on datasets that had not been used to select those cut points. Accordingly, uncertainty from model stochasticity was represented through repeated-seed dispersion, while cross-dataset generalization was assessed through held-out graphs rather than through an assumed population sample size.

The implementation used Python-based scientific computing tools, including PyTorch and PyTorch Geometric for graph neural-network computation, NetworkX for graph operations, and NumPy, pandas, SciPy, and scikit-learn for numerical, statistical, and downstream evaluation procedures. The repository did, however, centralize model parameters, dataset definitions, splits, graph variants, and reproducibility seeds in its configuration and experiment modules.

4 Experiments and Results

The final analysis employed the GraphSAGE with K=10, the structural feature set and the five graph variants originally on which the decision rules were based: original, psi, degree, centrality and hybrid. Later two more conditional strategy analyzers have been added, both of which are hybrid namely Hybrid Centrality and Hybrid Degree. Results were highly dependent on the task. Structural augmentation emerged as a winning strategy in nine of 19 cases out of which 6 were in controlled environment, and four as ties where structural augmentation gives identical results to original and six cases in which original graph remains better. Conversely, over 15 usable node-classification cells, augmentation resulted in no significant win but two ties and 13 times the original graph was better.

Table 1: Overall original-versus-augmentation evaluation.

Task	Augmentation better	Tie	Original better	Evaluated datasets
Link prediction	9 (47.4%)	4 (21.1%)	6 (31.6%)	19
Node classification	0 (0.0%)	2 (13.3%)	13 (86.7%)	15

These results answered the main research question that structural augmentation was not a generally beneficial pre-processing step. The empirical boundary in node classification had been quite clear: there was no dataset evaluated that had been statistically improved by augmentation. In the case of link prediction, however, the practically equal distribution of useful and non-useful cases had warranted the learning of a decision process before the topology change. It then shifted from the

question of whether structural augmentation works, to when it should be used and what structural signal should be added. The control over the role of the encoder was maintained during the main comparison. As in its Neighbourhood-aggregation formulation, GraphSAGE had been used as a fixed message-passing model [2]. The structural graph, as opposed to the GNN structure, thus continued to be the key experimental factor.

4.1 Encoder Ablations and Configuration Selection

Prior to the cross-dataset decision study, various ablation studies were performed on Enzymes with the Ψ graph at $K=10$. The AUC for link prediction using direct virtual-edge positives had increased from 0.6632 (walk derived positives) to 0.6909, with node-classification F1 score remaining almost the same. Then, mean aggregation outperformed weighted, max, and sum aggregations for both tasks. The depth experiment had offered the best separation of architectures - two layers had improved to 0.6909 AUC, three layers were down to about 0.4983, which shows increasing layers can increase noise.

Table 2: Encoder ablations used to fix the GraphSAGE configuration.

Ablation	Variant	NC weighted F1	LP AUC	Decision
Positive pairs	Walk co-occurrence	0.5405 ± 0.0014	0.6632 ± 0.0114	–
Positive pairs	Direct edges	0.5413 ± 0.0011	0.6909 ± 0.0156	Selected
Aggregation	Mean	0.5413 ± 0.0011	0.6909 ± 0.0156	Selected
Aggregation	Weighted mean	0.5399 ± 0.0020	0.6528 ± 0.0240	–
Aggregation	Max	0.5284 ± 0.0041	0.6378 ± 0.0353	–
Aggregation	Sum	0.5114 ± 0.0017	0.5063 ± 0.0596	–
Depth	1 layer	0.4971 ± 0.0018	0.6678 ± 0.0075	–
Depth	2 layers	0.5413 ± 0.0011	0.6909 ± 0.0156	Selected
Depth	3 layers	0.4506 ± 0.0187	0.4983 ± 0.0407	–

The GraphSAGE settings were tested as we have modified it for the new setting: structural virtual graphs instead of the traditional graph neighbourhoods, and using features that are purely structural, and an unsupervised embedding objective. So, it was necessary to tune one encoder configuration before comparing graph variants, otherwise a result could be induced by the tuning of the encoder and not by the construction of the graphs.

Later, the feature ablation revealed that none of the structural features had been the most important in all of the domains. The best results for node classification were found in clustering alone on Enzymes and Proteins, and in degree plus centrality compared to the four-feature combination, on Cora and Citeseer. The highest achieved centrality was 0.6498 on Enzymes, and degree plus centrality 0.5914 on Proteins in link prediction. The molecular graphs of random and constant controls were also typically significantly less strong. The results gave evidence that the four structural inputs of the common experimental pipeline should be retained, not only be considered individually in each of the datasets.

4.2 Stage-one Rules Discovery

The following experiments had characterized and connected those properties to the noticed gap between the original graph and best augmentation. Following the motivation of [5], adjusted homophily had been used rather than raw homophily because it accounted for the class-dependent chance baseline. Seven primary graph properties had been screened on the seven-dataset discovery panel. Candidate thresholds had been checked using the relative performance gap, a fixed-variant gap that removed max-selection bias, leave-one-dataset-out stability, and leave-one-out threshold prediction.

Based upon these results shown in Table 3, we can identify the patterns that high homophily prefers keeping the original graph, low homophily favors augmentation, while a very large connected component tends to favor augmentation and vice versa. Table 3, lists the various properties of the graph tested as possible Stage-one decision rules.

The **Fit Interval** represents the range in threshold values for which the separation between “augment” and “keep original” cases would be the same. For instance, adjusted homophily is specified

Table 3: Candidate Stage-one rules.

Candidate	Decision rule	Fit interval	Spearman ρ	LOO	Baseline	Final status
Adjusted homophily	Augment if $< \mathbf{0.227}$	0.093–0.361	−0.900/ − 1.000	$\mathbf{4/5 = 0.800}$	0.600	Lead Stage-one Rule
Largest-component fraction	Augment if $> \mathbf{0.959}$	0.918–1.000	+0.780/ + 0.780	$\mathbf{5/6 = 0.830}$	0.670	Unlabelled fallback
Neighbour predictability	Augment if < 0.408	0.332–0.484	−0.700/ − 0.600	$3/5 = 0.600$	0.600	Rejected
Original-retention gate*	Augment if $< \mathbf{0.012}$	0.006–0.018	−0.857 [†]	$\mathbf{6/7 = 0.857}$	0.714	Low-homophily gate

*Applied within the low-homophily cases. [†]Spearman correlation reported for the fitted gate.

as a threshold of 0.227; any threshold value of 0.093 to 0.361 would yield the same fitted decision-pattern. The wider the interval, the less sensitive to small changes would be the exact threshold.

The **Spearman correlation** indicates the degree of the correlation between each graph property and the observed augmentation benefit. A value of near +1 or −1 means there is a strong relationship, and a near zero value means it is a weak relationship. The sign indicates the direction. In the case of the adjusted homophily, the negative value indicates that as homophily increases, augmentation becomes less useful.

The **LOO (leave-one-out)** test aims at stability by removing one dataset to fit the rule on the remaining datasets, and to examine whether the removed data set can be correctly predicted. For instance, $4/5 = 0.800$ translates to four out of five cases accurately predicted.

The **Baseline** refers to the accuracy that can be achieved by just selecting the most common answer without applying the graph property. A useful rule should have some better results of LOO than this baseline. Lastly, the Final status indicates whether the candidate ended up becoming the primary rule, the fallback/gate rule or they were rejected.

Original-Retention Gate: Adjusted homophily provides the first Stage-one decision. A high value suggests that the original topology should be kept. However, a low adjusted-homophily value does not always mean that augmentation will help. Therefore, a second measure, called the *original-retention gate*, is used only inside this low-homophily region. Original retention measures how many edges from the original graph are also present in the Ψ -based structural-role graph. It is defined as

$$R_{\text{ret}} = \frac{|E_{\text{original}} \cap E_{\text{virtual}}|}{|E_{\text{original}}|}, \quad (13)$$

where E_{original} is the set of original edges and E_{virtual} is the set of edges in the virtual graph.

A high value means that the structural-role graph preserves more of the original connectivity, whereas a low value means that the rewiring creates a substantially different topology. The fitted Stage-one gate is

$$R_{\text{ret}} < 0.012 \implies \text{Augment}, \quad (14)$$

otherwise the decision inside this region is to keep the original graph. The threshold was fitted within the low-homophily region and should therefore be treated as a fitted decision rule rather than a universally validated boundary.

4.3 Prospective Validation on Unseen Datasets

We chose a truly separate validation or held-out set. These rules were not created on these new held-out datasets, which included Citeseer-LINQS, Proteins, PubMed, Actor, Minesweeper, Amazon Photo, Lastfm Asia, Amazon Ratings, and Squirrel-filtered. Each Rule-1 and Rule-2 decision had been written before the first encoder run performed on all these datasets, thus, isolating the Rule Discovery Process from the Rule Evaluation Process. Usually, the calculation of Rule 1 was 5/7 and for Rule 2, it was 4/7, and PubMed and Amazon Photo were not scored since observed differences were in the seed-noise band. An important pattern had been asymmetric. Whereas high adjusted homophily had transferred moderately well - suggesting that augmentation should benefit - low adjusted homophily had not ensured that it would - as the Minesweeper and Amazon Ratings data clearly showed.

Table 4: Stage-one rule predictions and observed link-prediction outcomes on the validation datasets.

Dataset	Adj. h	Largest comp. frac.	R1 prediction	R2 prediction	Predicted LP verdict	Actual LP verdict	Original	Best aug.
citeseer_linqs	0.6731	0.6464	Keep original	Keep original	Keep original	Keep original	0.6218	0.5437
proteins	0.3552	0.0143	Keep original	Keep original	Keep original	Keep original	0.6720	0.5834
pubmed	0.6860	1.0000	Keep original	Augment	Keep original (R1)	Tie	0.6392	0.6238
actor	0.0028	1.0000	Augment	Augment	Augment	Augment	0.5953	0.6617
minesweeper	0.0094	1.0000	Augment	Augment	Augment	Keep original	0.7093	0.6671
amazon_photo	0.7850	0.9787	Keep original	Augment	Keep original (R1)	Tie	0.7857	0.7803
lastfm_asia	0.8562	1.0000	Keep original	Augment	Keep original (R1)	Keep original	0.7218	0.7002
amazon_ratings	0.1402	1.0000	Augment	Augment	Augment	Keep original	0.7536	0.6550
squirrel_filtered	0.0086	1.0000	Augment	Augment	Augment	Augment	0.7399	0.7736

Table 4 experiments had then exposed the main limitation. We placed an original_retention gate in the low-homophily area, to separate out cases like “Minesweeper” and “Squirrel-filtered”. It had a frozen cutoff of 0.0119. Low adjusted homophily had therefore behaved as a useful warning that augmentation might be worth considering, but it had not been sufficient to determine the decision. The fitting analysis had achieved 6/7 leave-one-out decisions, but this gate still required genuinely unseen validation.

4.4 Stage-two signal selection

Stage-two does not recommend adding a structural signal until the augmentation was proposed in Stage-one. In this ten-seed strategy experiment, 14 datasets of labelled link prediction were used. 7 of these showed augmentation results and thus directly contributed to the signal selection process. Neighbour-label predictability had produced the only retained primary rule: when adjusted neighbour predictability surpassed 0.0092, the experiment chose centrality-based augmentation; below this threshold, it went with degree or with Ψ .

Table 5: Stage-two signal-selection results on augmenting datasets.

Dataset	Adj. neighbour predictability	Observed winner(s)	Stage-two decision
tolokers	-0.3274	Ψ / degree	Ψ or degree, unresolved
questions	-0.0041	degree	Ψ or degree, unresolved
actor	0.0060	degree	Ψ or degree, unresolved
squirrel_filtered	0.0123	centrality / degree	centrality
roman_empire	0.3324	centrality	centrality
pubmed	0.6946	centrality	centrality
lastfm_asia	0.8498	centrality	centrality

The ordering above presented in Table 5 presents a good reason for the choice of the threshold: the three observations that fell below the boundary showed that they had been won by Ψ or degree, while all the observations above the boundary included centrality amongst the winners. At the end of Stage 2, the team should have six datasets, instead of just the three LINKX graphs for which Stage-two was actually completed. Earlier both the Texas and Chameleon-filtered had been in previous tests and blocked by Stage-one.

The final Stage-two presentation shown in Table 6 had included six datasets rather than only the three LINKX graphs on which Stage-two actually ran. Chameleon-filtered and Texas had belonged the same tests and had been stopped by Stage-one. Amherst41, Johns Hopkins55, and Cornell5 proceeded to Stage-two and were all assigned centrality before their encoders were trained.

Table 6: Stage-one and Stage-two evaluation across the test datasets.

Dataset	Stage 1	Stage 2	Prediction	Centrality	Second-best	Interpretation
				score	result	
chameleon_filtered	KEEP	No	N/A	—	—	No augmentation suggested
texas	KEEP	No	N/A	—	—	No augmentation suggested
reed98	KEEP	No	N/A	—	—	No augmentation suggested
amherst41	AUGMENT	Yes	Centrality	0.6749	Hybrid-cent. 0.6718	Centrality ranked first
johnshopkins55	AUGMENT	Yes	Centrality	0.7160	Hybrid-cent. 0.7021	Centrality ranked first
cornell5	AUGMENT	Yes	Centrality	0.7078	Hybrid 0.7050	Centrality ranked first; small margin

When the Stage-two hypothesis was tested on the three held-out datasets, *amherst41*, *johnshopkins55*, and *cornell5*, the centrality-based virtual graph achieved the highest mean link-prediction score on all three datasets. However, the strength of this result differed across the datasets. On *johnshopkins55*, centrality showed the clearest advantage, reaching 0.7160 compared with 0.7021 for the second-best hybrid-centrality variant. On *amherst41*, centrality ranked first with 0.6749, while hybrid-centrality followed closely with 0.6718. Since both variants use the centrality signal, this result still supports the prediction that centrality is the appropriate structural signal for both datasets. For *cornell5*, centrality also ranked first with 0.7078, but the hybrid variant was very close at 0.7050. Therefore, this case supports centrality by ranking, but does not provide strong evidence that the pure centrality graph is clearly better than the hybrid alternative.

5 Discussions

5.1 Discussion

The results show that structural augmentation should not be treated as an automatic improvement over the original graph. Across the seven graph variants, the relative performance of the original, role-based, and hybrid graphs changed across datasets. This suggests that the usefulness of an added structural connection depends on how much useful information is already present in the original topology. When local Neighbours already provide the relationships needed for node classification, replacing or extending those edges may add little value and may even introduce connections that are less useful for the task. Adding or replacing edges to nodes may not be of great benefit, if relationships are already present in Neighbourhood areas. After all, when nodes that play similar roles are widely separated in the original graph or when they are not directly connected, structural augmentation can be useful. This result aligns with the principle of GraphSAGE, which is based on aggregating information from the sampled Neighbourhoods of nodes, thereby altering the information that is available during the message-passing process.

The comparison between Ψ , degree, and eigenvector centrality also shows that a more detailed structural measure does not necessarily provide the best graph in every case. The Ψ signature combines degree-distribution information, eigenvector centrality, KL-style divergence, and a Poisson formulation, whereas degree and centrality describe simpler aspects of node structure. Identity2Vec showed that combining structural properties through a Poisson-based identity measure can capture nodes with similar structural roles, including nodes that are not close in the original network. Our findings support the general idea that structural identity contains useful information, but they also show that the usefulness of a particular structural signal depends on the graph. In some cases, a simple measure such as degree or centrality can already contain enough information to form useful role-based Neighbourhoods. Therefore, the added complexity of Ψ should not be assumed to produce better classification by itself.

The contribution is therefore not another random-walk embedding method, but an examination of whether structural-role similarity can be placed directly into the topology seen by a GNN. The results also make an important distinction between replacing the original topology and extending it. Where hybrid variants perform better, the likely explanation is that they preserve useful physical relationships while adding access to structurally similar nodes. Where they do not improve performance, the additional edges may simply introduce less relevant messages into the aggregation process. This helps explain why there is no reason to expect one augmented graph to be best for every network.

5.2 Limitations

Several limitations should be considered when interpreting these findings. First, the final experiments use a fixed $K = 10$ for structural Neighbours. Although this provides a consistent comparison between graph variants, the same Neighbourhood size may not be suitable for graphs with very different sizes, densities, or degree distributions. A fixed K can add relatively few edges to one graph but substantially change another.

Second, the study focuses on GraphSAGE with a fixed aggregation setting. Since GraphSAGE explicitly builds representations from local Neighbourhoods, it is a suitable model for studying topology changes, but the observed effects cannot yet be assumed to hold for GIN, GAT, or other message-passing architectures. GraphSAGE itself was designed around Neighbourhood sampling and aggregation, meaning that its behavior is closely tied to the Neighbourhood structure provided to the model.

Finally, eigenvector centrality is normalized separately within connected components, so nodes from different components may receive similar values because they occupy similar relative positions within their own components, even when those components have very different structures.

5.3 Future Work

First, future work should involve studying the possibility of structuring augmentation in an adaptive manner as opposed to the same way for all graphs. The value of K might get selected from a set of graph properties, rather than being fixed at $K = 10$. The value of K could have been selected from one of the following: (i) density, (ii) component structure, (iii) degree variation, or (iv) distribution of structural-signature distances. A node specific K would be better because it would let the nodes with clear structural matches to get more role edges without getting extra role edges for the other nodes whose original Neighbourhoods are already informative.

A second approach is to untangle the effect of structural features from the effect of structural rewiring by selectively ablating the features. Further, degree-based augmentation is worth testing with and without degree as an input feature, and the degree and the centrality augmentations ought to be tested with and without centrality as an input feature, respectively, and the same with respect to Ψ . This would give increased confidence that it is not just a repetition of accessing the same signal, but that the new topology is responsible for any additional gain.

One more direction is to use the Centrality rule as a starting point to expand the strategy selection stage 2 framework. Future research must explore and validate other rules which may be more appropriate than Centrality when deciding which strategy to adopt in the cases of Ψ and Degree and their hybrid combinations. Lastly, other GNN architectures need to be evaluated on the other seven variations of the graph with more diverse graph types. Future research might learn, however, which situations require keeping the graph as is, which ones call for adding additional nodes, and whether Ψ or degree or centrality or a mixture is optimal.

6 Conclusion

The principal goal of this study was to break away from the one-size-fits-all paradigm of structure-aware graph augmentation and to introduce a characterization-guided decision framework. We investigated the potential of using a measurable structural property of the original graph to make a decision regarding which virtual graph construction is better suited to train the GNN before it is applied to the graph. Structural information in the form of Poisson similarity score, degree, eigenvector centrality and their combinations have been used to create six structure-based virtual graph variants evaluated using GraphSAGE. The primary result was influenced by the graph in which the augmentation occurred, not by the augmentation itself. That is, for no case was a virtual graph always the best option. The combination of the experiments had produced more selective answers than a universal graph-rewiring. For node classification, the evidence had suggested that it was best to keep the original topology, as none of the 15 tested datasets had a significant augmentation gain. For link prediction, the 6 augmentation cases clearly won, 6 original-graph wins and 4 ties had shown augmentation was helpful. High adjusted homophily had given evidence to support the retention of the original graph, and low adjusted homophily had only found a region where augmentation was still

possible. The second stage had yielded encouraging centrality-selection pattern: If a node’s neighbourhood label pattern is clearly and consistently indicative of its class means when Neighbour-label predictability is higher, our study suggests that the augmentation with eigenvector-centrality seems to work best.

The resulting contribution had therefore been a two-stage characterization-guided decision-making process, with a strong empirical boundary and very few examples or evidence linked to an augmentation of nodes, mixed but nevertheless substantial evidence of benefits of link prediction on structural augmentation, a useful but not perfect Stage-1 screen, and a promising Stage-2 centrality rule that needed to be validated for both their assertiveness and their negativity. This directly solved the original problem where nodes that are structurally similar could need new connections for messages, but it also quantitatively showed those connections shouldn’t be present by default. Thus, characterization-guided augmentation may evolve into a concrete decision framework where graph structure guides the augmentation applied prior to the start of GNN learning.

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